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COMPARISON OF SLEEP ESTIMATION USING APPLE WATCH ACCELEROMETRY AGAINST POLYSOMNOGRAPHY

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Introduction: Consumer wearables offer new ways to improve our health and well-being, including sleep. Researchers are interested in consumer wearables because their widespread adoption creates the potential for larger studies than could be run with clinically validated measurement methods, as those are more expensive or less convenient. This study investigates sleep tracking using sensor data from Apple Watch in comparison to the gold standard polysomnography (PSG).
Method: We used Apple Watch accelerometer data to establish both activity and heart rate (using ballistocardiography). Thirty participants (13 female, 17 male) wore the Apple Watch on their non-dominant wrist during clinical PSG. We compared predicted sleep status at the epoch level and overall sleep parameters, taking PSG as the ground truth.
Results: Our method achieved sleep-wake classification accuracy of 84%, sensitivity of 95%, and specificity of 47%. Apple Watch overestimated total sleep time (mean+SD) by 39.4 + 57.7 mins, underestimated WASO by 45.5 + 54.6 mins and the number of awakenings by 5.0 + 6.9. We observed worse performance for participants who had PSGs exhibiting frequent respiratory events.
Discussion: Accelerometry cannot replace PSG for diagnostic purposes. However, the Apple Watch results compare favourably to previously published Actiwatch-PSG comparisons. The performance we measured suggests that Apple Watch based accelerometry could be used in longitudinal studies to gather information similar to clinically validated accelerometers, potentially on a larger scale for lower cost. Further study is needed to understand how sleep disorders affect this kind of measurement.

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ACCURACY AND RELIABILITY OF THE TRANSCUTANEOUS CARBON DIOXIDE (TCCO2) SIGNAL IN THE SLEEP LABORATORY

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Carbon Dioxide (CO_2) monitoring is an essential part of assessing and treating disorders of hypoventilation in the sleep laboratory. While reliablity issues have been previously reported with the Transcutaneous Carbon Dioxide ($TcCO_2$) signal, there is limited data assessing the validity of this signal or its trend in the sleep laboratory context. Therefore, this study aimed to investigate the change in $TcCO_2$ accuracy from the beginning to the end of the sleep study in real world conditions across two different Victorian public hospital sleep laboratories that used two different $TcCO_2$ monitors. The sample included 13 consecutive patients from Monash Health and 44 consecutive patients from Alfred Health with an average age of 64 and 56 years respectively. Arterial Blood Gas (ABG) measurements were taken prior to and following each sleep study and compared concurrently with the $TcCO_2$ value. Bland-Altman analysis revealed an average difference between $TcCO_2$ and $PaCO_2$ of 3.29mmHg with agreement between -11.44 and 16.64mmHg for the TCM4 device and 1.31mmHg with agreement between -7.64 and 9.05mmHg for the TCM5 device. When accuracy was compared across time points for each patient, 46% of patients had an overnight accuracy change of ≥ 8mmHg when using the TCM4 compared with 20% when using the TCM5. It was concluded that the $TcCO_2$ signal was un-reliable across the different monitors and that the $TcCO_2$ trend may be difficult to interpret with confidence without blood gas calibration at the commencement and conclusion of the sleep study.

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CHRONOTYPE AND OSA COMBINE TO MODIFY RISK OF HYPERTENSION

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Introduction: There are limited data on the association of chronotype and hypertension and on their interaction on hypertension. This study aimed to investigate the independent and combined effects of chronotype and OSA on risk for prevalent hypertension in a middle-aged community population.
Methods: Baseline data on adult participants (n=1098, female=58%; age mean [range]=56.7[40.8–80.6] years) from an Australian community cohort study were analysed. Shift workers and individuals with incomplete data were excluded. Prevalent hypertension was defined as 'doctor diagnosed' and/or an elevated average systolic blood pressure (BP; ≥140mmHg) or diastolic BP (≥90mmHg). OSA was diagnosed when apnoea hypopnoea index (AHI) ≥10 events/hour from in-laboratory polysomnography. Chronotype was determined from actigraphy mid-sleep time on work free days. Tertiles of mid-sleep time were used to categorise morning, intermediate and evening chronotypes. Logistic regression (adjusted for sex, body mass index, age, alcohol consumption and sleep duration) were used to assess the cross-sectional relationship between chronotype, OSA and hypertension.
Results: After applying exclusion criteria 496 participants were analysed (female=58%; age mean[range]=57.0[42.1–81.6] years). All those with OSA had greater odds of hypertension than those without and there was no difference in risk of hypertension according to chronotype. Compared to morning chronotypes with no OSA (n=84), evening chronotypes with OSA (n=79) had non-significantly increased odds (OR 2.15, 95% CI 1.00–4.76; P=0.054) for hypertension while morning chronotypes with OSA (n=82) had significantly increased odds (OR 3.02, 95% CI 1.44–6.58; P=0.004).
Discussion: Morning chronotypes with OSA might be at increased risk of hypertension compared to evening chronotypes with OSA.